VS-05 Statistical Product Composition

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VS-05 Statistical Product Composition

1. Purpose and value created

VS-05 turns the statistics that VS-04 produced into finished statistical products. A single estimate is not yet something a user reads. VS-05 brings results from different domains, surveys, and time periods together into coherent publications, time series, indices, and datasets, explains them, and approves them for release. It does not re-estimate anything: it selects, combines, checks, explains, and signs off. The value created is **coherence and interpretability**: the product is internally consistent, coherent with related and prior statistics, and accompanied by the metadata and interpretation that users need to understand it.

VS-05 corresponds to the business domain **Statistically composed and composite products** (see *Steady-State Architecture, Business domains*). This section describes the business layer of that domain for all four process types: Design, Implementation, Execution, and Management. Re-estimation, weighting, and model-based analysis are not part of VS-05; they belong to VS-04. Publishing, distribution, user services, and final disclosure control for release belong to VS-06.

> **i** Signature discipline: coherence and interpretability
>
> VS-05 composes validated outputs; it does not re-estimate them. Every product is internally consistent, coherent with related and prior statistics, carries the accuracy metadata of its components, has lineage to its component outputs and the composition method, and is approved against dissemination and confidentiality standards before it is registered.

PLACEHOLDER: ArchiMate view VS-05 (a) value stream and capabilities

2. Position and state transition

VS-05 is the fifth primary value stream. It receives validated statistics from VS-04 and delivers approved products to VS-06. It is the stage where separate analyses become finished statistical products.

VS-05 works within State #4. It consumes analytical versions of #4 and produces approved versions of #4. Approval does not change an analytical version: it creates a new, immutable version with the status *approved* that refers to the analytical versions it was composed from. The quality gate, the minimum metadata set, and the protection of State #4 are described in *State #4: Statistics* and are not repeated here.

VS-05 receives the analytical versions together with the metadata VS-04 attached: methodology, weights, accuracy measures such as sampling errors and confidence intervals, and the record of the disclosure control already applied. VS-05 propagates this accuracy and quality metadata into the product. It does not discard or recompute it. It uses it to weight or caveat the composed figures, and to run coherence checks across the combined outputs. Composition never silently changes a component estimate.

From	To	Gate	Event out
#4 Statistics (analytical)	#4 Statistics (approved)	Approval gate of State #4 (component selection, composition, consistency and coherence, interpretive content, dissemination and confidentiality standards, sign-off)	*Statistical Product Registered*

PLACEHOLDER: ArchiMate view VS-05 (c) state handover and events: analytical #4 versions from VS-04, the composition service at the boundary (integrate outputs, construct composites, coherence checks, interpret, approve and register), approved #4 product versions handed to VS-06

3. Stages

The stages describe the value that VS-05 adds in every production cycle. Together they form the execution process type of VS-05. In a mature steady state, the products, composites, checks, and templates have been designed and built in advance (see attribute 4). Execution then focuses on composing the product, checking its coherence, and capturing component lineage and approval, so that every product is coherent, interpretable, and reproducible.

Stage	Name	Value added
VS-05.1	Select and Integrate Source Datasets	Relevant results are brought together
VS-05.2	Construct Composite Indicators or Indices	Indices, ratios, and aggregates are available
VS-05.3	Perform Consistency and Coherence Checks	The product is internally consistent
VS-05.4	Generate Analytical and Interpretive Outputs	Results are explained and visualised
VS-05.5	Approve and Register Statistical Products	The product is approved for release (State #4, approved)

Each stage below is described with its activities. The column *SAF process* shows the business process from *Business Processes* that the activity belongs to. The column *Other reference* shows the principles, capabilities, or external frameworks that the activity applies.

VS-05.1 Select and Integrate Source Datasets

Value created: *The right validated outputs brought together as the basis for a coherent product.*

Execution starts when VS-04 publishes the *Output Data Registered* event for the components of a product. The team applies the designed selection rules to pick the correct analytical #4 versions, by domain, period, and accuracy. The exact versions are pinned, so the product can always be traced to them. Components from several domains, surveys, or time periods are integrated into the product structure as specified. Their accuracy and methodology metadata is carried forward, and the component lineage is recorded. Registered outputs are reused; VS-05 does not repeat analysis that already exists, in line with SAF-P08.

Activity	Key outputs	SAF process	Other reference
Select and pin the correct component output versions according to the specification	Selected components	PRD-6.1 Prepare draft outputs; DLM-6 Provide data for reuse	SAF-P08
Integrate the components into the product structure	Integrated product base	PRD-5.1 Integrate data; PRD-6.1 Prepare draft outputs	1.3.6 Integrate statistics
Carry component accuracy and methodology metadata into the product	Propagated metadata	OVA-2 Metadata Management	SAF-P09
Record the component lineage: which output versions were used	Component lineage	OVA-2 Metadata Management	CSDA Traceability

VS-05.2 Construct Composite Indicators or Indices

Value created: *Derived measures, such as indices, ratios, and time series, that summarise the results and add meaning.*

This stage computes the designed composite indicators, indices, and time-series aggregates from the integrated components. Indicators and indices are calculated with the designed formulas and weighting. Time series are chained, linked, and benchmarked according to the specification. Each composite carries an uncertainty measure derived from the accuracy of its components. No component estimate is re-estimated: composites are built from the validated outputs as received.

Activity	Key outputs	SAF process	Other reference
Calculate composite indicators and indices with the designed formulas	Composite indicators	PRD-5.7 Calculate aggregates; PRD-5.5 Derive new variables and units	
Construct time series: chain, link, benchmark	Composite time series	PRD-5.7 Calculate aggregates	
Derive composite uncertainty from component accuracy	Composite accuracy measures	PRD-5.7 Calculate aggregates	SAF-P09
Record the composition method and inputs as lineage	Composite lineage	OVA-2 Metadata Management	SAF-P09

VS-05.3 Perform Consistency and Coherence Checks

Value created: *Confidence that the product is internally consistent and fits the wider statistical picture.*

This stage runs the designed checks. Internal consistency checks confirm that totals reconcile to their components and that breakdowns sum to totals. Coherence checks compare the product with related

statistics and with prior releases, within the designed tolerances. Each discrepancy is dispositioned according to the policy: accepted, flagged, or returned for review. A significant discrepancy is explained. Nothing is adjusted silently. If a component is wrong, it goes back to VS-04 and returns as a new analytical version. All check results are logged in the audit trail of the product.

Activity	Key outputs	SAF process	Other reference
Run internal consistency checks: totals and breakdowns reconcile	Consistency results	PRD-6.2 Validate outputs	
Run coherence checks against related statistics and prior releases	Coherence results	PRD-6.2 Validate outputs	1.3.6 Integrate statistics
Disposition discrepancies (accept, flag, return to VS-04) and explain them	Discrepancy dispositions	PRD-6.2 Validate outputs; DLM-2 Promote data through a quality gate	
Log all check results in the audit trail of the product	Audit trail entries	OVA-4 Process Data Management	SAF-P10

PLACEHOLDER: VS-05 composition and coherence architecture: component outputs of several domains integrated and composed into indicators, then checked for internal consistency and for coherence, with prior-period products as comparison, resulting in a coherent product that carries its lineage and accuracy

VS-05.4 Generate Analytical and Interpretive Outputs

Value created: *Products that users can understand: narrative, key messages, and visualisations grounded in the data.*

This stage produces the content that makes the product understandable. The analytical narrative and commentary are written with the designed templates. Key messages are stated together with their supporting evidence. Tables and visualisations are produced. The content passes the editorial and quality review, which checks that it is accurate, in plain language, accessible, and consistent with the data and its caveats, including the accuracy carried from VS-04. The review makes sure that the text does not overstate a movement that is not statistically significant.

Activity	Key outputs	SAF process	Other reference
Write the analytical narrative and commentary with the templates	Product narrative	PRD-6.3 Interpret and explain outputs	1.4.1 Statistical (output) analysis
State key messages with their supporting evidence	Key messages	PRD-6.3 Interpret and explain outputs	
Produce tables and visualisations	Tables and visualisations	PRD-7.2 Produce dissemination products	
Run the editorial and quality review of the content	Reviewed content	PRD-7.2 Produce dissemination products; OVA-1 Quality Management	GAMSO Quality Management

VS-05.5 Approve and Register Statistical Products

Value created: *A validated, approved, and registered product, ready for dissemination.*

This stage validates, approves, and registers the product. The product is checked against the dissemination and confidentiality standards. This includes the disclosure risk that arises from combining outputs: two tables that are each safe can together reveal a value. The product is then submitted to the designated approval authority. On approval, a persistent, immutable product version is created with a unique identifier and full lineage: components, composition method, coherence results, and approval record. Its metadata is registered in line with SAF-P09, and access permissions are applied in line with SAF-P10. The stage ends by publishing the *Statistical Product Registered* event, which tells VS-06 that an approved product is available for dissemination. If the product is not approved, it is revised, or a component is returned to VS-04.

Output: Approved statistical product (State #4, approved), the official handover artefact to the dissemination domain. Its minimum metadata set is described in *State #4: Statistics*.

Activity	Key outputs	SAF process	Other reference
Validate the product against dissemination and confidentiality standards, including the risk of combining outputs	Validation result	PRD-6.4 Apply disclosure control; PRD-6.5 Finalise outputs	SAF-P11; 1.3.7 Statistical disclosure
Submit the product to the approval authority and obtain sign-off	Approval record	PRD-6.5 Finalise outputs	4.1.5 Accountability
Create an immutable, identified product version with full lineage	Versioned product	DLM-3 Register and catalogue data; DLM-5 Store and version data	CSDA Traceability
Apply access permissions to the approved version	Access-controlled product	DLM-4 Classify and protect data	SAF-P10
Register metadata and publish the *Statistical Product Registered* event to notify VS-06	Catalogue registration, event	DLM-3 Register and catalogue data	SAF-P07

Each product version is linked to four records, all traceable to the analytical #4 versions it was composed from.

Product record	Content
Statistical product	Persistent identifier, immutable version, product type (publication, index, time series, dataset), reference period, registration timestamp
Component lineage (one per component)	Analytical #4 version used, domain or source, composition method, accuracy carried
Coherence result	Internally consistent, coherent with related statistics, coherent over time, number of flagged discrepancies
Interpretive content	Narrative reference, key messages with evidence, tables and charts, editorial review passed
Approval record	Approval authority, confidentiality cleared, approval timestamp, cleared for dissemination

4. The four process types

The sub-processes below are specific to VS-05. Each refines one or more SAF business processes, listed as *Refines*, which keeps it traceable to *Business Processes*. Sub-processes marked *(standard)* work as described in Standard activities in every value stream; only the VS-05 specifics are listed.

Design: provides metadata

Design sub-process	Refines	Main outputs
Design the Statistical Product Portfolio	PRD-2.1 Design outputs and dissemination; PRD-2.5 Design processing and analysis; DLM-1 Plan and design data assets; OVA-2 Metadata Management	Product specification, component-selection specification, metadata-propagation rules, product metadata standard
Design Composite Indicators and Indices	PRD-2.5 Design processing and analysis; OVA-2 Metadata Management	Composite-indicator specification, time-series specification, uncertainty-representation rules
Design Consistency and Coherence Checks	PRD-2.5 Design processing and analysis; DLM-7 Revise data; OVA-1 Quality Management	Consistency and coherence check sets, coherence tolerance policy, revision policy
Design Analytical and Interpretive Content	PRD-2.1 Design outputs and dissemination; OVA-1 Quality Management	Content templates, visualisation specification, key-message guidelines, editorial review policy
Design Approval, Confidentiality, and Versioning Framework	PRD-2.5 Design processing and analysis; PRD-2.6 Design production systems and workflows; DLM-1 Plan and design data assets; DLM-4 Classify and protect data	Approval policy, confidentiality standard, versioning and lineage policy, access control design

Design the Statistical Product Portfolio

Refines: PRD-2.1 Design outputs and dissemination; PRD-2.5 Design processing and analysis; DLM-1 Plan and design data assets; OVA-2 Metadata Management.

The starting point is deciding which products will be produced. This sub-process specifies each statistical product: its purpose, target users, content, structure, breakdowns, release frequency, and the analytical #4 outputs it draws on. The team designs how component outputs are selected and integrated (which estimates, which domains, which time periods) and how their accuracy metadata is carried into the product. Each product also gets its classification and versioning identity. Without product specifications, composition becomes ad hoc, and Management has no baseline to measure coherence and interpretability against.

Activity	Key outputs	SAF process	Other reference
Specify each product: purpose, users, content, structure, breakdowns, frequency	Product specification	PRD-2.1 Design outputs and dissemination	1.1.2 Design statistics product
Define the component outputs each product draws on, and the selection rules	Component-selection specification	PRD-2.5 Design processing and analysis	SAF-P08
Define how component accuracy and quality metadata is carried into the product	Metadata-propagation rules	OVA-2 Metadata Management	SAF-P09

Activity	Key outputs	SAF process	Other reference
Define the product classification and versioning identity	Product metadata standard	DLM-1 Plan and design data assets; OVA-2 Metadata Management	SAF-P09

Design Composite Indicators and Indices

Refines: PRD-2.5 Design processing and analysis; OVA-2 Metadata Management.

This sub-process specifies the derived measures that the products present: composite indicators, indices, ratios, and time-series aggregates. Each is defined by its formula, component inputs, weighting or aggregation method, and base period. Time series get chaining, linking, and benchmarking rules. Each composite gets the metadata and classification it carries. Where a composite combines outputs of differing accuracy, the design specifies how its uncertainty is represented. International standards, such as index manuals, inform the specifications.

Activity	Key outputs	SAF process	Other reference
Specify composite indicators and indices: formula, inputs, weighting, base period	Composite-indicator specification	PRD-2.5 Design processing and analysis	Index manuals
Specify time-series construction: chaining, linking, benchmarking	Time-series specification	PRD-2.5 Design processing and analysis	
Define how component uncertainty is represented in composites	Uncertainty-representation rules	PRD-2.5 Design processing and analysis	SAF-P09
Define the metadata and classification of each composite	Composite metadata standard	OVA-2 Metadata Management	SAF-P09

Design Consistency and Coherence Checks

Refines: PRD-2.5 Design processing and analysis; DLM-7 Revise data; OVA-1 Quality Management.

This sub-process designs the checks that make sure integrated products are sound. It specifies internal consistency checks: totals reconcile to components, and breakdowns sum to totals. It specifies coherence checks against related statistics, such as the national accounts and prior products, and across time. Tolerance thresholds and disposition rules (accept, flag, return for review) complete the check set. The revision policy for products is also designed here.

Activity	Key outputs	SAF process	Other reference
Define internal consistency checks: totals and breakdowns reconcile	Consistency check set	PRD-2.5 Design processing and analysis	
Define coherence checks against related statistics and across time	Coherence check set	PRD-2.5 Design processing and analysis; OVA-1 Quality Management	GAMSO Quality Management
Define tolerance thresholds and disposition rules	Coherence tolerance policy	PRD-2.5 Design processing and analysis	
Define the product revision policy	Revision policy	DLM-7 Revise data; OVA-1 Quality Management	GAMSO Quality Management

Design Analytical and Interpretive Content

Refines: PRD-2.1 Design outputs and dissemination; OVA-1 Quality Management.

This sub-process designs the content that makes products interpretable for people. It specifies the structure of the analytical narrative and commentary, the visualisations and tables, the key messages and the evidence they require, and the editorial and quality review of the content. Accessibility and plain-language standards are part of the design.

Activity	Key outputs	SAF process	Other reference
Specify the structure of the analytical narrative and commentary	Content template	PRD-2.1 Design outputs and dissemination	
Specify visualisations and tables	Visualisation specification	PRD-2.1 Design outputs and dissemination	
Define key messages and their evidence requirements	Key-message guidelines	PRD-2.1 Design outputs and dissemination	
Define the editorial and quality review of content	Editorial review policy	OVA-1 Quality Management	GAMSO Quality Management

Design Approval, Confidentiality, and Versioning Framework

Refines: PRD-2.5 Design processing and analysis; PRD-2.6 Design production systems and workflows; DLM-1 Plan and design data assets; DLM-4 Classify and protect data.

The last design sub-process makes sure that products are properly approved, protected, and traceable. It defines the approval workflow and the approval authority. It defines the confidentiality and disclosure standards that a product must meet before registration, including the risk of combining outputs. It sets the product versioning policy (identity, revision triggers, immutability, retention) and the lineage required from each product to its component outputs. Role-based access control for products is designed in line with SAF-P10.

Activity	Key outputs	SAF process	Other reference
Define the approval workflow and the approval authority	Approval policy	PRD-2.6 Design production systems and workflows	4.1.5 Accountability
Define confidentiality and disclosure standards for products, including the risk of combining outputs	Confidentiality standard	PRD-2.5 Design processing and analysis	SAF-P11; 3.3.1 Privacy management
Define product versioning and lineage requirements	Versioning and lineage policy	DLM-1 Plan and design data assets	CSDA Traceability
Design role-based access control for products	Access control design	DLM-4 Classify and protect data	SAF-P10; 3.3.4 Logical identity and access control

PLACEHOLDER: VS-05 design process decomposition: inputs (analytical #4 outputs with accuracy, user needs, framework standards, product and index standards such as SDMX and index manuals), the five design sub-processes in order, and the design artefact each produces, which together govern Implementation, Execution, and Management

Implementation: provides the execution process

Implementation sub-process	Refines	Main outputs
Build Integration and Composition Engines	PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows	Configured integration and composition engines, pinned connections to the State #4 catalogue
Build Indicator-Construction and Time-Series Engines	PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows	Indicator-construction engine, time-series engine, composite-accuracy component
Build Coherence-Check, Content, and Approval Tooling	PRD-3.2 Reuse or build processing and analysis components; PRD-3.3 Reuse or build dissemination components; PRD-3.4 Configure workflows; DLM-5 Store and version data	Coherence tooling, content tooling, approval and versioning store, event publication
Test and Validate the Pipeline *(standard)*	PRD-3.5 Test production systems; PRD-3.6 Test statistical business process; PRD-3.7 Finalise production systems	Test reports; integration test from output intake through composition, coherence checks, content, and approval to registration; coherence test: composites reconcile to components and cohere with prior products; reproducibility test: a product regenerates identically from its component versions plus the composition method

Build Integration and Composition Engines

Refines: PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows.

This sub-process turns the product and integration designs into operational tools. The integration engine selects and combines the designed component outputs into the product structure, with their accuracy and quality metadata. The composition engine assembles products from their components according to the specifications. Connections to the State #4 catalogue use version pinning, so the exact component versions used are recorded. All composition logic is version-controlled, in line with SAF-P10.

Activity	Key outputs	SAF process	Other reference
Build or configure the integration engine: select and combine component outputs	Configured integration engine	PRD-3.2 Reuse or build processing and analysis components	1.1.3 Build statistics product
Build or configure the composition engine: assemble products from components	Configured composition engine	PRD-3.2 Reuse or build processing and analysis components	
Configure pinned connections to the State #4 output catalogue	Output-source integration	PRD-3.4 Configure workflows	CSDA Metadata Management
Put all composition logic under version control	Versioned composition repository	PRD-3.4 Configure workflows	SAF-P10

Build Indicator-Construction and Time-Series Engines

Refines: PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows.

The designed composite-indicator and time-series specifications are implemented. Indicator and index calculators are built from the designed formulas and weighting. Time-series engines are configured for chaining, linking, and benchmarking. Uncertainty representation is implemented, so that composites carry an appropriate accuracy measure. Every composite output is produced with its method and inputs attached.

Activity	Key outputs	SAF process	Other reference
Build indicator and index calculators from the designed formulas	Indicator-construction engine	PRD-3.2 Reuse or build processing and analysis components	
Configure time-series chaining, linking, and benchmarking	Time-series engine	PRD-3.2 Reuse or build processing and analysis components	
Implement uncertainty representation for composites	Composite-accuracy component	PRD-3.2 Reuse or build processing and analysis components	SAF-P09
Attach the method and inputs to every composite	Composite metadata binding	PRD-3.4 Configure workflows	SAF-P09

Build Coherence-Check, Content, and Approval Tooling

Refines: PRD-3.2 Reuse or build processing and analysis components; PRD-3.3 Reuse or build dissemination components; PRD-3.4 Configure workflows; DLM-5 Store and version data.

This sub-process implements the coherence checks, the content tooling, and the approval and versioning store. Coherence tooling implements the designed internal consistency checks and the checks across sources and time, with their tolerances. Content tooling supports the analytical narrative, tables, and visualisations according to the templates, with the editorial review workflow. The approval and versioning store makes each approved product version immutable, identified, reproducible from its components plus method, and accompanied by its lineage and approval record. Event notifications are configured to alert VS-06 when a product is approved and registered.

Activity	Key outputs	SAF process	Other reference
Implement consistency and coherence checks with their tolerances	Coherence tooling	PRD-3.2 Reuse or build processing and analysis components	
Build content and visualisation tooling with the editorial workflow	Content tooling	PRD-3.3 Reuse or build dissemination components	
Build the approval and versioning store: immutable, identified, reproducible	Approval and versioning store	PRD-3.4 Configure workflows; DLM-5 Store and version data	CSDA Traceability
Configure the *Statistical Product Registered* event for downstream consumers	Event publication	PRD-3.4 Configure workflows	SAF-P07

PLACEHOLDER: VS-05 implementation pipeline: the five design artefacts feeding the three build sub-processes, followed by technical testing, coherence test, and reproducibility test, resulting in an operational composition pipeline and operational procedures

Execution: provides the statistical data

Execution runs the stages of VS-05 for every product release. The stages and their activities are described in attribute 3. A cycle starts when the analytical versions a product needs have been registered, and it is planned against the release calendar. Execution applies the designs and the solution as they are; a new composite, check, or template goes back to Design. A coherence problem is never solved inside VS-05 by changing a component: the component goes back to VS-04.

Aspect	VS-05
Trigger	*Output Data Registered* for the components of a product
Cadence	Per product release, according to the release calendar
Performed by	ROL-10 Data engineer (integration), ROL-15 Subject-matter statistician, ROL-17 Release manager (approval authority), ROL-03 Data steward
Input	Registered analytical #4 versions and their metadata
Output	Registered approved #4 version with composition record, coherence results, interpretive content, and approval record
Control point	Coherence checks in VS-05.3; approval gate of State #4 in VS-05.5
Exception path	Component that fails the coherence check returned to VS-04 for a new analytical version; product not approved revised and resubmitted; need for a new composite or check back to Design

Management: provides quality reports

Management sub-process	Refines	Main outputs
Monitor Process Performance *(standard)*	OVA-4 Process Data Management; PRD-8.1 Gather evaluation inputs	Process performance dashboard; throughput report against the release calendar; paradata on component versions and methods used and on approval iterations
Assess Product Quality	OVA-1 Quality Management; DLM-8 Monitor data quality, lineage and use	Coherence report, interpretability review, product completeness report, quality trend analysis
Manage Product and Version Lifecycle	OVA-2 Metadata Management; DLM-5 Store and version data; DLM-8 Monitor data quality, lineage and use	Product version register, metadata quality report, version store audit, metadata lineage audit

Management sub-process	Refines	Main outputs
Audit and Compliance Reporting *(standard)*	OVA-1 Quality Management; OVA-4 Process Data Management; DLM-8 Monitor data quality, lineage and use	Reproducibility audit: every product regenerates from its component versions plus the composition method (4.1.5 Accountability); approval and confidentiality report confirming that every registered product was approved by the designated authority and meets the confidentiality standard, including the risk of combining outputs (SAF-P11); standards compliance report on product and exchange standards, such as SDMX; security compliance report on access controls and the integrity of the approval record (SAF-P10)
Continuous Improvement *(standard)*	PRD-8.2 Conduct evaluation; PRD-8.3 Agree an action plan	Improvement recommendations, for example refined composite-indicator methods, coherence checks that produce fewer false alarms, a streamlined approval workflow, better visualisations, and more reuse of validated outputs; trends watched: coherence issues, approval iterations, component reuse, interpretability

Assess Product Quality

Refines: OVA-1 Quality Management; DLM-8 Monitor data quality, lineage and use.

Beyond process performance, the quality of the products themselves is evaluated against the criteria set in Design. Each product is assessed on four dimensions. Coherence covers internal consistency and consistency with related and prior statistics. Interpretability covers the clarity and accuracy of narrative and visualisations. Completeness checks that all designed content is present. Accuracy propagation checks that component accuracy was carried correctly. Quality indicators are produced per product and per cycle, and trends are tracked to detect recurring coherence issues or interpretive weaknesses. This follows the Data Quality capability of CSDA and SAF capability 2.1.3 *Statistical quality management*.

Activity	Key outputs	SAF process	Other reference
Evaluate coherence: internal, and with related and prior statistics	Coherence report (per cycle)	OVA-1 Quality Management	2.1.3 Statistical quality management
Assess the interpretability of narrative and visualisations	Interpretability review	OVA-1 Quality Management	CSDA Data Quality
Check completeness and correct accuracy propagation	Product completeness report	DLM-8 Monitor data quality, lineage and use	2.1.3 Statistical quality management

Activity	Key outputs	SAF process	Other reference
Identify quality trends across products and cycles	Quality trend analysis	DLM-8 Monitor data quality, lineage and use	2.1.3 Statistical quality management

Manage Product and Version Lifecycle

Refines: OVA-2 Metadata Management; DLM-5 Store and version data; DLM-8 Monitor data quality, lineage and use.

The product portfolio and the versions it produces must stay current, complete, and well governed. This sub-process governs the life cycle of product specifications, composite-indicator definitions, and content templates. It versions them, manages transitions such as the rebasing of an index, and retires obsolete products. It verifies that every product carries full component lineage, composition method, coherence results, and approval record, as a compliance check on SAF-P09. It governs the product version store: retention, immutability, and reconciliation of catalogue entries with stored versions. Metadata lineage from outputs through composition to products is tracked.

Activity	Key outputs	SAF process	Other reference
Govern the life cycle and versions of products, indicators, and templates	Product version register	OVA-2 Metadata Management	SAF-P09
Verify metadata completeness: component lineage, method, coherence results, approval	Metadata quality report	OVA-2 Metadata Management	SAF-P09
Govern the product version store: retention, immutability, reconciliation	Version store audit	DLM-5 Store and version data	CSDA Traceability
Track metadata lineage from outputs to products	Metadata lineage audit	DLM-8 Monitor data quality, lineage and use	CSDA Traceability

PLACEHOLDER: VS-05 management feedback loop: data and paradata from Execution flowing through the five management sub-processes (coherence pass rate and timeliness, coherence and interpretability, products and versions, reproducibility and approval), with dashboards to the process owner, quality reports to the quality manager, compliance reports to VS-07, and improvement recommendations back to Design

Metadata per process type

In this domain, metadata is what makes a composite product interpretable: each product must carry its components, methodology, coherence results, and approval. The table shows the data metadata each process type of VS-05 contributes; the paradata are listed in Standard activities in every value stream. In Execution, VS-05 also records the approval workflow trail and the number of approval iterations as paradata, and in Implementation the outcomes of the coherence tests.

Process type	Data metadata produced
Design	Product specifications, composite-indicator definitions, coherence criteria, approval and confidentiality policy

Process type	Data metadata produced
Implementation	Integration configurations, indicator-construction parameters, coherence-check parameters, product templates
Execution	Component lineage, composition and indicator method, coherence-check results, interpretive content, and approval record per product
Management	Coherence and interpretability assessments, product quality scores, lineage assessments, trend analyses

PLACEHOLDER: VS-05 business object lifecycle: design artefacts govern implementation assets (integration and composition engines, indicator engines, coherence and content tooling, approval store), which produce the process that composes analytical #4 versions into approved statistical products, generating component lineage, coherence results, and approval records that management assesses and feeds back to design

5. Actors and roles

VS-05 is mostly internal: it receives its input from VS-04, and the process owner of VS-04 remains the data owner of the analytical versions. The table shows the actors and the reference roles involved. Specialised role names used in an NSI map onto these reference roles (see *Role names in the value streams*); for example, the product manager and the coherence designer in the original VS-05 description are ROL-06 Statistical designer, and the approval authority is ROL-17 Release manager. Design work is typically carried out by the collaboration COL-05 *Statistical product team*.

Actor or role	ID	Process type	Responsibility in VS-05
International statistical organisation (standards body)	ACT-07	External	Maintains the product, indicator, and exchange standards the NSI conforms to, such as SDMX and index manuals
Process owner	ROL-01	Management	Accountable for the end-to-end performance of VS-05
Data owner	ROL-02	Management	Process owner of Product Composition; accountable for the coherence and interpretability of approved #4 versions
Data steward	ROL-03	Execution	Registers approved products and versions, manages metadata entries, applies access controls
Data custodian	ROL-04	Execution	Operates the product version store and keeps the pre-release access log

Actor or role	ID	Process type	Responsibility in VS-05
Statistical designer	ROL-06	Design	Designs the product portfolio, product specifications, and release content (product manager); designs the consistency and coherence checks (coherence designer)
Methodologist	ROL-07	Design	Specifies composite indicators, indices, and time-series construction methods (indicator and index methodologist)
Data engineer	ROL-10	Implementation, Execution	Builds and configures the integration, indicator-construction, and reporting engines; in each cycle, selects and integrates the component outputs into the product structure (integration specialist)
Platform engineer	ROL-11	Implementation	Configures composition services, output-source connections, and access controls
Test manager	ROL-12	Implementation	Plans and carries out testing of the composition pipeline: technical, coherence, and reproducibility tests
Subject-matter statistician	ROL-15	Execution	Produces the analytical and interpretive content: commentary, key messages, and visualisations (subject-matter author)
Disclosure control specialist	ROL-16	Design, Execution	Defines and checks the confidentiality standard for products, including the disclosure risk of combining outputs
Release manager (approval authority)	ROL-17	Execution	Validates products against the dissemination and confidentiality standards and approves them for registration and release
Quality manager	ROL-18	Management	Monitors coherence and interpretability indicators, produces quality reports, and drives continuous improvement

Actor or role	ID	Process type	Responsibility in VS-05
Compliance officer	ROL-19	Management	Ensures confidentiality and standards compliance of products and produces audit reports

6. Capabilities, processes, and functions

Capability foundations

The table shows which capabilities from the SAF capability model VS-05 relies on, and what role they play. NSIs can use it to map the value stream to their own organisation.

Capability group	Capabilities	Role in VS-05
1.1 Statistical design and build	1.1.2 Design statistics product, 1.1.3 Build statistics product	Design the statistical products and composite indicators, and build the composition pipeline
1.3 Processing	1.3.6 Integrate statistics, 1.3.7 Statistical disclosure	Integrate separate statistics into a coherent product, and check the disclosure risk of the combined product
1.4 Analysis	1.4.1 Statistical (output) analysis	Validate, interpret, and explain the product, and finalise it for dissemination
2.1 Data management	2.1.1 Data lifecycle management, 2.1.2 Metadata management, 2.1.3 Statistical quality management, 2.1.4 Statistical register management	Manage products, metadata, and quality; assess coherence; integrate across registers and time
3.2 Computerisation and automation	3.2.1 Application development, 3.2.2 Functional management	Build and maintain the integration, indicator-construction, and reporting and visualisation engines
3.3 Security and privacy	3.3.1 Privacy management, 3.3.2 Security planning	Validate products against confidentiality standards and assess the disclosure risk of composite products
4.1 Governance	4.1.1 Strategy and governance, 4.1.2 Policy and planning, 4.1.4 Improvement management, 4.1.5 Accountability	Govern the domain, steward product approval, plan improvements, report on compliance

Stages, processes, and functions

The table shows which SAF business processes realise each stage of VS-05 and which business functions they mainly use.

Stage	Realised by processes	Main functions
VS-05.1 Select and Integrate Source Datasets	PRD-5.1 Integrate data; PRD-6.1 Prepare draft outputs	FUN-3.4 Estimate and analyse
VS-05.2 Construct Composite Indicators or Indices	PRD-5.5 Derive new variables and units; PRD-5.7 Calculate aggregates	FUN-3.3 Process statistical data; FUN-3.4 Estimate and analyse

Stage	Realised by processes	Main functions
VS-05.3 Perform Consistency and Coherence Checks	PRD-6.2 Validate outputs	FUN-3.4 Estimate and analyse; FUN-5.2 Assure and review quality
VS-05.4 Generate Analytical and Interpretive Outputs	PRD-6.3 Interpret and explain outputs; PRD-7.2 Produce dissemination products	FUN-3.4 Estimate and analyse; FUN-3.6 Disseminate statistics
VS-05.5 Approve and Register Statistical Products	PRD-6.4 Apply disclosure control; PRD-6.5 Finalise outputs; DLM-3 Register and catalogue data	FUN-3.4 Estimate and analyse; FUN-3.5 Protect statistical confidentiality; FUN-7.2 Manage metadata

PLACEHOLDER: ArchiMate view VS-05 (b) stages realised by processes and functions

Framework mappings

This reference table maps each process type of VS-05 to the SAF business processes, the international frameworks (GSBPM, GAMSO, CSDA), and the SAF capability model. It is intended for architects and auditors; the rest of this section does not depend on it.

Process type	SAF business processes	GSBPM	GAMSO	CSDA	SAF capabilities
Design	PRD-2.1, PRD-2.5, PRD-2.6, DLM-1, DLM-4, DLM-7, OVA-1, OVA-2	Phase 2 Design (2.1, 2.5, 2.6)	Production (design activities), Capability Management	Data Governance, Metadata Management, Data Quality	1.1.2 Design statistics product, 2.1.2 Metadata management, 2.1.3 Statistical quality management
Implementation	PRD-3.2 to PRD-3.7, DLM-5	Phase 3 Build (3.2 to 3.7)	Capability Management	Data Processing and Dissemination preparation (set-up), Data Governance (configuration)	1.1.3 Build statistics product, 3.2.1 Application development, 3.2.2 Functional management
Execution	PRD-5.1, PRD-5.5, PRD-5.7, PRD-6.1 to PRD-6.5, PRD-7.2, DLM-2 to DLM-6, OVA-2, OVA-4	Phase 6 Analyse (6.1 to 6.5), 7.2 Produce dissemination products; the business domain table also lists 7.1 Update output systems	Production (execution)	Data Analysis and Dissemination preparation (execution)	1.3.6 Integrate statistics, 1.4.1 Statistical (output) analysis, 2.1.2 Metadata management, 2.1.3 Statistical quality management

Process type	SAF business processes	GSBPM	GAMSO	CSDA	SAF capabilities
Management	OVA-1, OVA-2, OVA-4, DLM-5, DLM-8, PRD-8.1 to PRD-8.3	Phase 8 Evaluate; overarching Quality Management and Metadata Management	Corporate Support (Quality, Data, and Metadata Management), Capability Management (improvement)	Data Quality, Data Governance, Traceability	2.1.3 Statistical quality management, 4.1.4 Improvement management, 4.1.5 Accountability

7. Principles

The following principles govern VS-05 most directly.

Principle	Statement	Application in VS-05
SAF-P01	Actively participate in open-source communities to maintain collective control over statistical processes	Composition, indicator-construction, and reporting and visualisation tooling is reused and developed in open-source communities with other NSIs, so that the NSI keeps collective control over critical product capabilities instead of depending on vendors.
SAF-P07	Implement loosely coupled processes and systems	Composition is independent from upstream estimation and downstream dissemination. The approved statistical product is the decoupling interface: VS-06 uses it without knowing how it was composed.
SAF-P08	Data is shared property	Approved products and their components, once registered, are available to all authorised consumers. Composition reuses existing validated outputs instead of duplicating analysis.
SAF-P09	No data without metadata and classification	Every product carries its component outputs and their accuracy, the composition and indicator method, the coherence results, its classification, and the approval record. This is what makes products interpretable and trustworthy.
SAF-P10	Security By Design	Access controls, integrity verification, and a tamper-evident composition and approval record are built into the pipeline, so that every product is attributable and reproducible.

Principle	Statement	Application in VS-05
SAF-P11	Privacy By Design	Products are validated against confidentiality standards. The disclosure risk of composite products, including the risk of combining outputs, is assessed by design before registration.

i Principles that apply across the process types

SAF-P09 underpins interpretability in all four process types: Design defines the product and its required metadata, Execution captures component lineage, composition method, and coherence results, and Management assesses product quality. SAF-P10 governs the tamper-evident composition and approval record across the stages VS-05.1 to VS-05.5. SAF-P11 applies especially to VS-05.5, where the confidentiality and disclosure risk of the composite product are confirmed before registration. SAF-P07 governs both the intake of analytical versions from VS-04 and the handover of approved versions to VS-06.

8. Interfaces, services, and events

Boundary interfaces

VS-05 has two boundary interfaces, both within State #4: intake of analytical versions from VS-04, and handover of approved versions to VS-06.

Boundary	Handed over	Required metadata	Trigger	Responsibility transfer
VS-04 → VS-05 (intake)	Registered analytical #4 version	Methodology and parameters, weights, accuracy measures, model diagnostics, applied disclosure control record, output version identifier, lineage to the processed source (#3), classification	*Output Data Registered*	VS-04 keeps custody of the analytical version; VS-05 takes responsibility for composition and approval on receipt of the event
VS-05 → VS-06 (handover)	Registered approved #4 version (statistical product)	Component lineage, composition and indicator method, propagated accuracy, coherence results, interpretive content, approval record, confidentiality status, product version identifier, classification	*Statistical Product Registered*	VS-05 keeps custody of the product version; VS-06 takes responsibility for final disclosure control, release, and user support on receipt of the event

Business services

Service	Offered to	Realised in	Interfaces
Integration and composition service	Composition teams, consumers of output data	VS-05.1 Select and Integrate Source Datasets	Pinned connections to the State #4 output catalogue
Indicator and index service	Composition teams, methodologists	VS-05.2 Construct Composite Indicators or Indices	Composite construction interface and a query interface for methodology and parameters
Coherence validation service	Quality managers, subject-matter statisticians	VS-05.3 Perform Consistency and Coherence Checks	Coherence results and discrepancy dispositions
Analytical content service	Authors, editors, VS-06 Dissemination and Service Delivery	VS-05.4 Generate Analytical and Interpretive Outputs	Content and visualisation tooling with editorial workflow
Product provisioning service	VS-06 Dissemination and Service Delivery	VS-05.5 Approve and Register Statistical Products	Event notification (*Statistical Product Registered*) and a catalogue query interface to find and access approved products, their lineage, and their accuracy
Product catalogue service	All data consumers in the NSI	Management sub-process *Manage Product and Version Lifecycle*	Search and browse interface for products, component lineage, coherence results, and approval records
Design and methodology service	Statistical designers, methodologists	All Design sub-processes of VS-05	Design documents, methodology advice

Business events

Event	Triggers or marks
Design artefacts published	Start of Implementation after a design change
Pipeline production-ready	End of Implementation; Execution can start
Output Data Registered	Start of VS-05.1 for the products that use that analytical version
Statistical Product Registered	Completion of VS-05.5 after approval; triggers VS-06. A product that is not approved is revised, or a component is returned to VS-04
Improvement recommendations issued	Input for Design from *Continuous Improvement*

PLACEHOLDER: VS-05 event choreography between VS-04, NSI teams, and process management across Design, Implementation, Execution (output data registered, integrate and compose, coherence checks and interpretive content, approval granted or refused), and Management (coherence and timeliness monitoring, product quality, audit reports, improvement recommendations)

Interaction with the supporting value streams

Value stream	Provides to VS-05	Receives from VS-05
VS-07 Governance and Compliance	Confidentiality and approval policies, revision policy framework	Reproducibility audits, approval and confidentiality compliance reports
VS-08 Innovation and Methodological Development	New composite-indicator and time-series methods	Improvement needs
VS-09 Infrastructure and Platform Enablement	Composition and content tooling, version store, catalogue, event interfaces	Capacity and tooling requirements
VS-10 Customer and Stakeholder Engagement	User needs for products and content	Product portfolio, feedback on interpretability

9. Key performance indicators

The key performance indicators are collected per production cycle in *Monitor Process Performance*. They feed the dashboard that the process owner reviews.

KPI	Description	Measurement
On-time product rate	Products composed within the release-calendar window	Per product, per cycle
Coherence-check pass rate	Products passing the coherence checks on the first run	Per product, per cycle
Approval iteration count	Number of approval cycles before sign-off	Per product, per cycle
Component reuse rate	Share of products that reuse existing validated outputs	Per cycle
Reproducibility rate	Share of products that regenerate from components plus method	Per cycle
Processing time	Time from output intake to product registration	Per product, per cycle

10. Examples

Labour market publication. The quarterly labour market publication is specified to combine the Labour Force Survey employment-rate estimates (national and NUTS-2), the unemployment series, and administrative job-vacancy figures into one coherent release, with the confidence interval of each component carried into the published tables. For Q3 2026, the team selects the validated `OD-LFS-2026Q3` employment and unemployment outputs and the latest administrative vacancy output, and pins their exact versions. The commentary explains a 0.3 percentage point rise in the employment rate, notes that it is within the confidence interval, and presents a regional map and a time-series chart. The editorial review confirms that the wording does not overstate a movement that is not statistically significant.

Business statistics product. The composite structural business statistics product integrates the register-based aggregates and the survey-based estimates for the same NACE by region domains. The engine pins the exact `OD-BUS` output versions and carries their accuracy metadata into the combined tables. In the coherence checks, NACE-class turnover sums to the published total, the survey-based and register-based components agree within tolerance for overlapping domains, and the series is coherent with the prior year. One domain exceeds the tolerance and is returned to VS-04 for review instead of being adjusted in the product.

Consumer price index. The headline consumer price index is composed from the validated item indices with the designed weighting and chaining. The all-items index and its main divisions are constructed, and each carries its method and the period weights used.

Toolkit: SAF-TK-V05 VS-05 maturity checklist and scenarios